

Pragalbha Patil

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Skills

Languages: Ruby, Python, Java, PHP, JavaScript, TypeScript, SQL

Frameworks & Libraries: Ruby on Rails, React, Next.js, Redux, Node.js, Express, NestJS, Spring Boot, Django, Flask, Laravel

Cloud & DevOps: AWS (EKS, ECS, EC2, S3, CloudFront, RDS, IAM, CloudWatch), GCP (GKE, Pub/Sub, BigQuery), Docker, Kubernetes, Terraform, Helm, GitHub Actions, Jenkins, NGINX, CDN

Data & Messaging: Databricks, Apache Spark (Delta), Kafka, RabbitMQ, ETL, Data Pipelines, Redis, Elasticsearch, ClickHouse

Databases: PostgreSQL, MySQL, MongoDB

APIs & Security: REST, GraphQL, WebSockets, Streaming API, OpenAPI/Swagger, OAuth2, OIDC, JWT, SSO, RBAC, Entitlements, Rate Limiting, Usage Metering

Observability & QA: OpenTelemetry, Prometheus, Grafana, ELK Stack, Datadog, Sentry, RSpec, Jest, pytest, Cypress, Playwright

Domain Expertise: Market Data Delivery, Nasdaq Data Link Platform, Fintech, Multi-Tenant SaaS, Data Onboarding, SLAs/SLOs

Experience

Senior Software Engineer, Nasdaq – Bangalore, IN May 2024 – Present

- Designed and implemented internal MCP server enabling AI models (LLMs) to interact with Nasdaq Data Link (data.nasdaq.com) APIs, Databricks, and internal data services
- Built table creation and dataset onboarding workflows on Databricks using NDL platform, improving research dataset availability and quality
- Developed automated market data ingestion and ETL pipelines, reducing client onboarding time by 30% and ensuring data validation and SLA compliance
- Enhanced Data Link API integrations (REST and WebSocket) for low-latency data delivery, adding entitlements, rate limiting, and usage metering controls
- Integrated Pendo analytics into Data Link web platform to track SDK (Python/Java/R) adoption, API usage patterns, and developer experience metrics
- Maintained and upgraded Ruby on Rails microservices for market data delivery, improving scalability, observability (OpenTelemetry, Prometheus, Grafana), and performance under high-throughput workloads

Senior Software Engineer, GoComet – Bangalore, IN June 2023 – May 2024

- Led enhancements to a multi-tenant analytics dashboard built with React/Redux, Node.js, and REST APIs, processing millions of trade records to deliver real-time P&L, market trend visualizations, and client-specific KPIs
- Optimized backend services in Node.js and Ruby on Rails, improving query performance and API response times by 25% through caching (Redis) and database indexing (PostgreSQL)
- Expanded authentication and authorization features using OAuth2, JWT, and role-based access controls, ensuring secure data access across web and API channels
- Delivered CI/CD improvements with GitHub Actions and Docker, reducing deployment time and improving release reliability
- Collaborated with product and data teams to integrate external APIs, standardize data ingestion pipelines, and ensure SLA compliance

Software Engineer, GoComet – Bangalore, IN July 2021 – June 2023

- Developed and maintained a high-volume analytics dashboard for supply chain and logistics insights, integrating

market data APIs and SQL data warehouses

- Built and refactored backend microservices (Ruby on Rails, Node.js) supporting millions of daily logins via web and API interfaces, with load balancing and horizontal scaling on AWS
- Implemented API documentation with OpenAPI/Swagger to streamline developer onboarding and reduce integration errors
- Introduced automated testing suites (RSpec, Jest) and monitoring (Prometheus, Grafana) to improve platform stability and observability
- Worked closely with stakeholders to gather requirements, design features, and iterate in Agile/Scrum sprints

Education

University of Mumbai - K.J. Somaiya, Bachelor of Engineering - BE, Information Technology August 2018 – July 2021

- CGPI: 8.38/10.0

• **Coursework:** Computer Architecture, Algorithms, Computational Theory, Machine Learning

Vivekanand Education Society's Polytechnic, High School Diploma in Computer Technology July 2015 – June 2018

- **Coursework:** Computer Architecture, Algorithms and Programming

Publications

Predictive Policing for Women's safety - SSRN June 2021
Vishwa Gosalia, *Pragalbha Patil*, Khushboo Mehta [dx.doi.org/10.2139/ssrn.3867200](https://doi.org/10.2139/ssrn.3867200)

Projects

Speech-to-American-Sign-Language Converter github.com/Pragalbha-Patil/S-TASLC-Python

- Built a Python-based system that converts live speech or typed input into American Sign Language, enabling accessible real-time communication for hearing-impaired users.
- Integrated NLP for speech recognition and gesture-mapping algorithms for sign output.
- Tech: Python, SpeechRecognition, OpenCV, NLP pipelines.

Multiplayer Battleship Game <https://playbattleship.online>

- Developed a real-time multiplayer Battleship game with WebSocket networking and server-side state management using Colyseus.
- Implemented matchmaking, turn-based logic, reconnection handling, and in-game chat.
- Tech: Unity, C#, Colyseus, WebGL, NGINX, AWS EC2.

Godot Puzzle-Platformer Game github.com/Pragalbha-Patil/first-godot-game

- Designed and developed a 2D puzzle-platformer game in Godot with custom physics, AI-driven enemies, and interactive level elements.
- Implemented save/load systems, procedural level generation, and polished UI/UX for smooth gameplay experience.
- Tech: Godot Engine (GDScript), Tilemaps, Pathfinding AI, Physics Engine.